HAORAN GUO

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EDUCATION

University of Science and Technology of China (USTC)

School of Information Science and Technology | Undergraduate | Sep 2024 - Expected Jun 2028

GPA: 3.52/4.3

Selected coursework: Mathematical Analysis, Linear Algebra, Modern Control Theory.

RESEARCH & ENGINEERING PROJECTS

NEXUS: 3D Shape Generation and Reconstruction Jul 15 - Aug 15, 2026

- A 3D mesh represents an object's surface with connected triangles. Implemented models based on NEXUS that learn vertex positions and their connections for 3D shape generation and reconstruction.
- Built PyTorch workflows for data processing, training, evaluation, and visualization; used controlled experiments to improve architecture and training choices.
- Achieved **99.63% surface reconstruction F1 on 100 fixed training shapes**, measuring the balance of recovered and extra faces. Evaluations under different posterior-noise conditions gave similar scores.

Persona-to-Scene: Multi-Agent Scene Generation and Editing Jul 15 - Aug 15, 2026

- Built collaborating AI agents that turn a person's interests and daily habits into a room layout and personalized interior images.
- Supported scene editing as activity needs change; used information tracking, instruction validation, and before/after image comparisons in end-to-end case studies.

SceneFun3D Track 1: Interactive 3D Scene Understanding

- Competed independently under the name Night's Watch, identifying interactive parts of indoor 3D scenes, such as handles and buttons.
- Completed environment and data preparation, baseline reproduction, training, and evaluation. Test-set results are listed on the official SceneFun3D benchmark.

RoboGame: RGB-D Perception and Robotic Grasping Jun - Oct 2026

- Built a vision-guided grasping system: a RealSense D435i depth camera locates colored blocks, and a robotic arm uses a suction cup to pick them up. Completed perception, arm control, and hardware integration testing.

Facial Expression Recognition

- Trained a lightweight MobileNetV2 classifier on FER2013 to recognize expressions such as happiness and surprise; compared several models on accuracy and computational cost.

TECHNICAL SKILLS

Programming: Python, C/C++, Linux/Ubuntu, Git.

Machine learning: PyTorch, training and debugging, evaluation, 3D mesh processing.

Robotics and tools: MuJoCo, STM32, MATLAB.

HONORS & ACTIVITIES

USTC Outstanding Student Scholarship; member of the USTC boxing team.